

Strategic Equity Analysis

5-Year Performance & Portfolio Strategy: Saudi Aramco vs. ExxonMobil

Executive Summary

This report synthesizes a comprehensive 5-year data engineering and financial analysis project comparing two global energy titans: Saudi Aramco (2222.SR) and ExxonMobil (XOM). By leveraging historical market data, feature engineering (Moving Averages, Daily Returns, Normalized Growth), and interactive Power BI visualizations, we have extracted actionable business intelligence. The core finding reveals a distinct divergence in asset behavior: Aramco operates as a highly stable, defensive anchor, whereas ExxonMobil acts as an aggressive growth engine.

1. Comparative Performance Insights

Based on the normalized growth algorithms and cumulative return metrics modeled over the past five years, the data reveals two fundamentally different investment profiles within the same macro-economic sector.

Saudi Aramco (The Defensive Anchor)

- **Market Behavior:** Aramco exhibited remarkably low volatility and resilient sideways-to-positive price action.
- **Return on Investment:** Over the 5-year period, the initial investment saw moderate, steady appreciation. While it did not yield exponential multi-bagger returns, its primary value proposition lies in its unshakeable stability.
- **Strategic Role:** Acts effectively as a "safe haven" equity. It is ideal for capital preservation, minimizing portfolio drawdowns, and providing steady foundational value during periods of market turbulence.

ExxonMobil (The Growth Engine)

- **Market Behavior:** XOM demonstrated significant bullish momentum with a steep upward growth trajectory and slightly higher volatility.
- **Return on Investment:** The normalized growth race definitively illustrates that ExxonMobil captured aggressive market upside, resulting in a substantial multiplier effect on the initial investment value.

- **Strategic Role:** Serves as the primary driver for capital appreciation. It is suited for risk-tolerant capital allocation aiming to drastically outpace inflation and maximize overall portfolio returns.

2. Correlation & Portfolio Diversification

One of the most critical discoveries of the Data Engineering phase was the calculation of the Pearson correlation coefficient between the daily returns of both assets.

Data-Driven Insight: Correlation Coefficient of -0.12

Despite both entities being heavily tethered to the global oil and energy markets, statistical analysis yielded a correlation of -0.12. This near-zero, slightly negative correlation indicates a statistically insignificant relationship between their day-to-day price movements.

Business Implication: Holding both stocks simultaneously does not double the sector risk. Instead, it creates an excellent opportunity for Portfolio Diversification. When one asset faces downward pressure, the other acts independently, effectively smoothing out the overall volatility curve of the investor's portfolio.

3. Final Investment Decision & Strategy

Strategic Recommendation

Based on the visualized risk-vs-return scatter plot and historical performance metrics, the optimal strategic decision is a bifurcated portfolio approach rather than an "either/or" selection:

- **For Aggressive Capital Growth:** Overweight the portfolio toward ExxonMobil (e.g., 60-70% allocation) to capture the steep upward momentum and maximize total Yield on Capital.
- **For Risk Mitigation & Preservation:** Allocate the remaining capital (e.g., 30-40%) to Saudi Aramco. Its low volatility metric ensures that a solid portion of the underlying capital is insulated from erratic market swings.

Conclusion: By merging the aggressive appreciation of ExxonMobil with the defensive stability of Saudi Aramco, an investor achieves a mathematically optimized balance of high returns and minimized systemic risk.